

Администрирование сетевых подсистем

Лабораторная работа №10 — Расширенные настройки SMTP-сервера

Новичков Максим

Российский
университет дружбы
народов, Москва,
Россия

Цели и задачи работы

Приобретение практических навыков по конфигурированию SMTP-сервера, настройке аутентификации пользователей и защите почтовых соединений с использованием TLS.

Выполнение лабораторной работы

Настройка LMTP в Dovecot

```
root@server:~ - sudo -i
dovecot.conf [----] 26 L:[ 14+11 25/103] *(1182/4322b) 0010 0x00A [*][X]
# protocol imap { }, local 127.0.0.1 { }, remote 10.0.0.0/8 { }

# Default values are shown for each setting, it's not required to uncomment
# those. These are exceptions to this though: No sections (e.g. namespace {})
# or plugin settings are added by default, they're listed only as examples.
# Paths are also just examples with the real defaults being based on configure
# options. The paths listed here are for configure --prefix=/usr
# --sysconfdir=/etc --localstatedir=/var

# Protocols we want to be serving.

protocols = imap pop3 lmtp

# A comma separated list of IPs or hosts where to listen in for connections..
# "*" listens in all IPv4 interfaces, "::" listens in all IPv6 interfaces.
# If you want to specify non-default ports or anything more complex,
# edit conf.d/master.conf.
#listen = *, ::

# Base directory where to store runtime data.
#base_dir = /var/run/dovecot/

# Name of this instance. In multi-instance setup doveadm and other commands
# can use -i <instance_name> to select which instance is used (an alternative
# to -c <config_path>). The instance name is also added to Dovecot processes
1 Help 2 Save 3 Mark 4 Replac 5 Copy 6 Move 7 Search 8 Delete 9 PullDn 10 Quit
```

Рис. 1: Настройка LMTP

Настройка SMTP-аутентификации

```
root@server:~ - sudo -i

10-master.conf [----] 15 L:[ 49+12 61/129] *(1537/3509b) 0010 0x00A [*][X]

inet_listener submission {
    #port = 587
}
inet_listener submissions {
    #port = 465
}
}

service lmtp {
    unix_listener /var/spool/postfix/private/dovecot-lmtp {
        group = postfix
        user = postfix
        mode = 0600
    }
}

service imap {
    # Most of the memory goes to mmap()ing files. You may need to increase this
    # limit if you have huge mailboxes.
    #vsz_limit = $default_vsz_limit

    # Max. number of IMAP processes (connections)
    #process_limit = 1024
}

1Help 2Save 3Mark 4Replac 5Copy 6Move 7Search 8Delete 9PullDn 10Quit
```

Рис. 2: SMTP-аутентификация

```
[root@server.manovichkov.net ~]# postconf -e 'smtpd_sasl_type = dovecot'
[root@server.manovichkov.net ~]# postconf -e 'smtpd_sasl_path = private/auth'
[root@server.manovichkov.net ~]# postconf -e 'smtpd_recipient_restrictions = reject_unknown_recipient_domain, permit_mynetworks, reject_non_fqdn_recipient, reject_unauth_destination, reject_unverified_recipient, permit'
[root@server.manovichkov.net ~]# postconf -e 'mynetworks = 127.0.0.0/8'
[root@server.manovichkov.net ~]# █
```

Рис. 3: Параметры Postfix

Настройка межсетевого экрана

```
root@client:~ - sudo -i

00a0 - e2 7f 5b a0 22 1f 81 13-0f e3 aa b1 a9 0f c5 f8  ..[.".....
00b0 - fb ff 98 a2 af 16 a2 8c-38 57 e9 5b c4 80 70 85  ....8W[..p.
00c0 - 07 b7 e2 7a b7 5d 8d 35-3e 6a 1b d5 f1 21 c8 93  ...z.].5>j...!..

Start Time: 1760695273
Timeout   : 7200 (sec)
Verify return code: 18 (self-signed certificate)
Extended master secret: no
Max Early Data: 0

---
read R BLOCK
EHLO test
250-server.manovichkov.net
250-PIPELINING
250-SIZE 10240000
250-VRFY
250-ETRN
250-AUTH PLAIN
250-ENHANCEDSTATUSCODES
250-8BITMIME
250-DSN
250-SMTPUTF8
250 CHUNKING
AUTH PLAIN bWfub3ZpY2hrb3YAbWfub3ZpY2hrb3YAMTIzNDU2
235 2.7.0 Authentication successful
quit
```


Account Editor

✕

Identity

Receiving Email

Receiving Options

Sending Email

Defaults

Composing Messages

Security

Server Type: **SMTP**

Description: For delivering mail by connecting to a remote mailhub using SMTP.

Configuration

Server: Port: ▼

☒ Server requires authentication

Security

Encryption method: ▼

Authentication

Type: ▼

Username:

Send Options

☐ Re-encode message before send

```
[root@server.manovichkov.net ~]# cd /vagrant/provision/server/
[root@server.manovichkov.net server]# cp -R /etc/dovecot/dovecot.conf /vagrant/provision/server/mail/etc/dovecot/
cp: overwrite '/vagrant/provision/server/mail/etc/dovecot/dovecot.conf'? y
[root@server.manovichkov.net server]#
[root@server.manovichkov.net server]# cp -R /etc/dovecot/conf.d/10-master.conf /vagrant/provision/server/mail/etc/dovecot/conf.d/
[root@server.manovichkov.net server]# cp -R /etc/dovecot/conf.d/10-auth.conf /vagrant/provision/server/mail/etc/dovecot/conf.d/
cp: overwrite '/vagrant/provision/server/mail/etc/dovecot/conf.d/10-auth.conf'? y
[root@server.manovichkov.net server]#
[root@server.manovichkov.net server]# mkdir -p /vagrant/provision/server/mail/etc/postfix
[root@server.manovichkov.net server]# cp -R /etc/postfix/master.cf /vagrant/provision/server/mail/etc/postfix/
[root@server.manovichkov.net server]# █
```

Рис. 6: Копирование конфигурации

Выводы по проделанной работе

В ходе выполнения лабораторной работы была настроена связка Postfix + Dovecot с поддержкой LMTP, SASL-аутентификации и TLS-шифрования.

Реализована безопасная передача сообщений, проверена работа через telnet и почтовый клиент.

Сервер готов к эксплуатации в защищённой среде и интегрирован в виртуальную инфраструктуру для автоматического развертывания.